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Debris bin assembly, cleaning robot and system thereof

Abstract

The present application provides a debris bin assembly including: a debris bin housing and a debris discharge valve; the debris bin housing includes an inner cavity, and an air inlet, a debris outlet, and at least one air guide portion that are disposed in communication with the inner cavity, and the air inlet is in communication with a debris inlet channel of the cleaning robot; the debris discharge valve is disposed at the debris outlet and configured to open the debris outlet to evacuate debris from the inner cavity in response to a vacuum negative pressure, and to close the debris outlet after the vacuum negative pressure is withdrawn; the at least one air guide portion is configured for being in communication with an external air and configured to guide a debris discharging airflow into the inner cavity in response to the vacuum negative pressure in the debris outlet.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) Pursuant to 35 U.S.C. § 119 and the Paris Convention, the present application claims the benefit of Chinese Patent Application No. 202111165421.6 filed Sep. 30, 2021, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present application relates to a field of cleaning robots, and more particularly to a debris bin assembly, a cleaning robot and a system thereof.

BACKGROUND

(3) While the cleaning robot (for example, a sweeping robot) automatically moves on the floor, the cleaning robot can suck the dust and debris on the floor into the debris bin of the robot through the air inlet channel to clean the area where the robot passes. Robots are fully developed and widely configured for the convenience.

(4) In order to avoid users from emptying debris bin frequently, a large-sized debris collector and a suction unit are added to the charging base in cooperation with the robot. When the cleaning robot returns to the charging base for charging, the suction unit sucks the debris from the debris bin of the robot into the debris collector of the charging base. This process is also called reverse vacuuming. At present, it is found that after reverse vacuuming, there is still a lot of debris left in the debris bin of the robot, and the debris discharging efficiency of the debris bin is lower.

SUMMARY

(5) Embodiments of the present application provide a debris bin assembly, a cleaning robot and a system thereof, so as to solve the technical problem that the dust left in the debris bin is more, and the debris discharging efficiency of the debris bin is lower.

(6) An embodiment of the present application provides a debris bin assembly, and the debris bin assembly is applied to a cleaning robot; the debris bin assembly includes: a debris bin housing and a debris discharge valve; the debris bin housing includes an inner cavity, an air inlet, a debris outlet, and at least one air guide portion; the air inlet, the debris outlet, and the air guide portion are disposed in communication with the inner cavity, and the air inlet is configured for being in communication with a debris inlet channel of the cleaning robot; the debris discharge valve is disposed at the debris outlet and configured to open the debris outlet to evacuate debris from the inner cavity in response to a vacuum negative pressure, and to close the debris outlet after the vacuum negative pressure is withdrawn; the at least one air guide portion is configured for being in communication with an external air and configured to guide a debris discharging airflow into the inner cavity in response to the vacuum negative pressure in the debris outlet.

(7) An embodiment of the present application provides a cleaning robot, and the cleaning robot includes a robot body and the debris bin assembly above-mentioned, the debris bin assembly is detachably mounted on the robot body.

(8) An embodiment of the present application further provides a cleaning robot system, and the cleaning robot system includes the cleaning robot above-mentioned and a cleaning base station; the cleaning base station is provided with a debris collection port configured to align with a debris outlet of the debris bin housing, and the cleaning base station sucks a debris in the inner cavity through the debris collection port and the debris outlet.

(9) Compared with the prior art, in the above-mentioned debris bin assembly, the cleaning robot and the system thereof, the debris bin housing is provided with the inner cavity, the air inlet, the debris outlet, and the at least one air guide portion; and the air inlet is configured for being in communication with a debris inlet channel of the cleaning robot; the at least one air guide portion is configured for being in communication with an external air and configured for responding to the vacuum negative pressure in the debris outlet to introducing debris discharging airflow to the inner cavity, therefore, the air volume of the debris discharging airflow can be increased through the at least one air guide portion, and the debris discharging airflow can pass through the at least one air

guide portion and the air inlet in different directions, which can increase the airflow at different positions inside the debris bin, and is beneficial to promote the residual debris in the inner cavity to be discharged through the debris outlet, thereby the debris evacuation efficiency of the debris bin assembly is improved and the problem of more debris remaining in the debris bin is solved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) One or more embodiments are exemplified by the figures in the corresponding drawings, and these exemplifications do not constitute limitations of the embodiments, and elements with the same reference numerals in the drawings are denoted as similar elements, unless otherwise stated, the figures in the accompanying drawings do not constitute a scale limitation.
- (2) FIG. 1 is a schematic diagram of a longitudinal cross-sectional structure of a cleaning robot provided by an embodiment of the present application;
- (3) FIG. 2 is a schematic view of an exploded structure of a robot body and a debris bin assembly of a cleaning robot provided by an embodiment of the present application;
- (4) FIG. 3 is a first schematic view of a cross-sectional structure of a debris bin assembly provided by an embodiment of the present application;
- (5) FIG. 4 is a second schematic view of a cross-sectional structure of a debris bin assembly provided by an embodiment of the present application;
- (6) FIG. 5 is a third schematic view of a cross-sectional structure of a debris bin assembly provided by an embodiment of the present application;
- (7) FIG. 6 is a fourth schematic view of a cross-sectional structure of a debris bin assembly provided by an embodiment of the present application;
- (8) FIG. 7 is a first schematic view of an exploded structure of a debris bin assembly provided by an embodiment of the present application;
- (9) FIG. 8 is a second schematic view of an exploded structure of the debris bin assembly provided by an embodiment of the present application;
- (10) FIG. 9 is a schematic view of an exploded structure of an air guide portion of a debris bin assembly provided in an embodiment of the present application;
- (11) FIG. 10 is a schematic view of a partial cross-sectional structure of a debris bin assembly provided by an embodiment of the present application; and
- (12) FIG. 11 is a schematic diagram of a longitudinal cross-sectional structure of a cleaning robot system provided by an embodiment of the present application.

DETAILED DESCRIPTION OF EMBODIMENTS

(13) In order to make the objectives, technical solutions and advantages of the present application clearer, the present application will be further described in detail below with reference to the accompanying drawings and embodiments. It should be understood that the specific embodiments described herein are only used to explain the present application, but not to limit the present application. Based on the embodiments of the present application, all other embodiments obtained by those skilled in the art without creative efforts shall fall within the protection scope of the present application.

(14) It should be noted that, if there is no conflict, various features in the embodiments of the present application may be combined with each other, which are all within the protection scope of the present application. In addition, although the functional modules are divided in the schematic view of the device, and the logical sequence is shown in the flowchart, in some cases, the modules in the device may be divided differently, or the sequence or the described steps shown in the flowchart may be performed. Furthermore, the words “first”, “second” and “third” used in the present application do not limit the data and execution order, but only distinguish the same or

similar items with basically the same function and effect.

(15) Referring to FIG. 1 to FIG. 4, an embodiment of the present application provides a cleaning robot **100**. The cleaning robot **100** includes a robot body **10** and a debris bin assembly **20** detachably connected to the robot body **10**.

(16) The debris bin assembly **20** includes a debris bin housing **21** and a debris discharge valve **22**, the debris bin housing **21** is provided with an inner cavity **211**, an air inlet **212**, a debris outlet **213**, and at least one air guide portion **214**, the air inlet **212**, the debris outlet **213** and the air guide portion **214** are arranged in communication with the inner cavity **211**, the air inlet **212** is arranged in communication with the debris inlet channel **30** of the cleaning robot **100**, and the debris discharge valve **22** is arranged at the debris outlet **213**, the debris discharge valve **22** is used to open the debris outlet **213** in response to the negative vacuum pressure to discharge the debris in the inner cavity **211**, and cover the debris outlet **213** after the negative vacuum pressure is withdrawn, the at least one air guide portion **214** is configured for being in communication with the external air, and the at least one air guide portion **214** is configured for introducing debris discharging airflow to the inner cavity **211** in response to the negative vacuum pressure at the debris outlet **213**.

(17) Compared with the prior art, in the above-mentioned debris bin assembly **20**, the debris bin housing **21** is provided with the inner cavity **211**, the air inlet **212**, the debris outlet **213**, and the at least one air guide portion **214**; and the air inlet **212** is configured for being in communication with a debris inlet channel **30** of the cleaning robot **100**; the at least one air guide portion **214** is configured for being in communication with an external air and configured for responding to the vacuum negative pressure in the debris outlet **213** to introducing debris discharging airflow to the inner cavity **211**, therefore, the air volume of the debris discharging airflow can be increased through the at least one air guide portion **214**, due to the different positions of the at least one air guide portion **214** and the air inlet **212**, and the debris discharging airflow can pass through the at least one air guide portion **214** and the air inlet **212** in different directions, and the debris discharging airflow can pass through the at least one air guide portion and the air inlet in different directions, which can increase the airflow at different positions inside the debris bin, and is beneficial to promote the residual debris in the inner cavity **211** to be discharged through the debris outlet **213**, thereby improving the debris evacuation efficiency of the debris bin assembly **20** and solving the problem of more debris remaining in the debris bin.

(18) It can be understood that the cleaning robot **100** can be any one of a sweeping robot, an integrated sweeping and mopping robot, a mopping robot or a washing floor robot. In other embodiment, the cleaning robot **100** may not be limited to the above examples.

(19) In the embodiment, the cleaning robot **100** is an integrated sweeping and mopping robot. The robot body **10** is a main body of the cleaning robot **100**, and the robot body **10** can be in any shape such as a circle, a rectangle, or a D shape, which is not limited herein.

(20) In an optional embodiment, the robot body **10** may also be of other design structures. For example, the robot body **10** is an integrally formed structure, and a left and right separated structure. The embodiment of the present application does not limit the material, shape, and structure of the robot body.

(21) The robot body **10** may include a chassis **11** and an upper cover assembly **12**. The upper cover assembly **12** is detachably mounted on the chassis **11** to protect various functional components inside the cleaning robot **100** from strongly impacting or damaging from accidentally dripping of spilled liquids during use; the chassis **11** and/or the upper cover assembly **12** are used to carry and support various functional components. The surface of the upper cover assembly **12** facing away from the chassis **11** forms an appearance surface, which can improve the overall appearance of the cleaning robot **100**, and buttons can be provided on the appearance surface to facilitate the user to operate the cleaning robot **100** through the buttons. A mounting cavity is formed between the chassis **11** and the upper cover assembly **12**, and the mounting cavity is used to provide an arrangement space for the internal components of the cleaning robot **100**. The cleaning robot **100**

can arrange a vacuum pump, a circuit board, a ground detection sensor, a collision detection sensor, and a wall sensor in the mounting cavity.

(22) The cleaning robot **100** includes a traveling mechanism **13** mounted on the chassis **11**. The traveling mechanism **13** includes two traveling wheels, at least one universal wheel, and a motor for driving the wheels to rotate. The two traveling wheels and the at least one universal wheel at least partially protrudes from the bottom portion **216** of the chassis **11**. For example, under the action of the weight of the cleaning robot **100**, the two traveling wheels can be partially hidden in the chassis **11**. In an optional embodiment, the traveling mechanism **13** may further include any one of a triangular crawler wheel, a Mecanum wheel, and the like. The traveling mechanism **13** may also not include the at least one universal wheel.

(23) The cleaning robot **100** may include at least one middle sweeping brush **14**, and the at least one middle sweeping brush **14** may be arranged in a receiving groove provided at the bottom portion of the chassis **11**, and a dust suction port is provided in the receiving groove, and the dust suction port is in communication with the debris bin assembly **20** and the dust suction fan, so that when the middle sweeping brush **14** rotates, dust and debris on the ground are stirred up, and the dust and debris are sucked into the debris bin assembly **20** from the dust suction port by using the suction force generated by the dust suction fan.

(24) The cleaning robot **100** can be designed to plan a path on the ground autonomously, or the cleaning robot **100** can be designed to move on the ground in response to remote control commands. The cleaning robot **100** can navigate through one or a combination of gyroscopes, accelerometers, cameras, GPS positioning, and/or laser radar, and so on, for example, the cleaning robot **100** can be provided with the laser radar protruding from the top surface, the surrounding environment is scanned by the laser radar to collect obstacle data, and an environment map is established based on the obstacle data, the real-time positioning can be performed according to the environment map, which is convenient for planning cleaning paths.

(25) In the embodiment, the debris bin assembly **20** is detachably mounted on the robot body **10**. The debris bin assembly **20** and the robot body **10** can be assembled in various forms. For example, in some embodiments, the chassis **11** of the robot body **10** is provided with a mounting groove on the peripheral side, and the debris bin housing **21** of the debris bin assembly **20** is detachably mounted in the mounting groove, the debris bin housing **21** can form the debris outlet **213** in the bottom portion **216** or the peripheral side, the debris bin housing **21** can be separately mounted in the mounting groove of the chassis **11**, the debris bin housing **21** can also be assembled with the water tank and then mounted in the mounting groove of the chassis **11**. Alternatively, in another embodiment, the upper cover assembly **12** of the robot body **10** is provided with a mounting groove, and the debris bin housing **21** of the debris bin assembly **20** is detachably mounted in the mounting groove, and the debris outlet **213** is formed at the bottom portion **216** of the debris bin housing **21**; the debris bin housing **21** can be separately mounted in the mounting groove of the upper cover assembly **12**, and the debris bin housing **21** can also be assembled with the water tank as a whole and then mounted in the mounting groove of the upper cover assembly **12**.

(26) The air inlet **212** of the debris bin housing **21** is in communication with the dust suction port on the chassis **11**. The debris bin housing **21** is further provided with an air outlet **219** in communication with the inner cavity **211**, and the debris bin housing **21** is in communication with the dust suction fan through the air outlet **219**. The debris bin assembly **20** further includes a filter **40**, and the filter **40** is connected to the debris bin housing **21**. The filter **40** is used to block the debris particles in the airflow from passing through the air outlet **219**, so as to trap the debris in the inner cavity **211** of the debris bin housing **21**. The filter **40** may be HEPA, sponge or other types of filter media.

(27) Referring to FIG. 1, FIG. 4, FIG. 10 and FIG. 11, the dust suction airflow can pass through the air inlet **212** and the debris outlet **213**, and after the external cleaning base station **200** is docked with the cleaning robot **100**, the cleaning base station **200** can generate a suction negative pressure

at the debris outlet **213** by a fan, and under the action of the negative pressure, the external air can form the dust suction airflow and pass through the air inlet **212** and the debris outlet **213** in turn. The air drives the debris to be discharged through the debris outlet **213** under the action of the suction negative pressure, so that the cleaning base station **200** can recover the debris stored in the debris bin housing **21**.

(28) Referring to FIG. **10**, the debris discharge valve **22** includes a cover member **221** and an elastic member. The cover member **221** is connected to the debris bin housing **21**, and the cover member **221** can move relative to the debris bin housing **21** to a position where covers or opens the debris outlet **213**. The cover member **221** can move relative to the debris bin housing **21** to a position where the debris outlet **213** is opened under the external suction effect, and after the suction effect is withdrawn, the cover member **221** is moved to a position where the debris outlet **213** is covered under the elastic recovery action of the elastic member or the action of its own elasticity. When the cover member **221** moves to the position where the debris outlet **213** is covered, the debris outlet **213** is in a closed state, and the cleaning robot **100** can normally clean the debris on the ground, and suck the debris into the debris bin housing **21** by suction force; when the cover member **221** moves to the position where the debris outlet **213** is opened, the debris outlet **213** is in an open state, and the cleaning base station **200** can evacuate the debris from box housing **21** through the debris outlet **213**. In other embodiments, the cover member **221** can also move relative to the debris bin housing **21** under the driving of a driving device. Alternatively, the cover member **221** may be an elastic material (eg, rubber or silicone), and the cover member **221** may automatically cover the debris outlet **213** under the action of its own elasticity, thereby saving additional elastic members.

(29) The cover member **221** may be any one of a plastic cover plate, a metal cover plate, a rubber cover plate, or a silicone cover plate.

(30) Referring to FIG. **1** to FIG. **4**, and FIG. **7**, in the embodiment, the debris bin housing **21** includes a top portion **215**, a bottom portion **216** disposed opposite to the top portion **215**, and a side shell **217** connected between the top portion **215** and the bottom portion **216**. The top portion **215**, the bottom portion **216**, and the side shell **217** are surrounded to form the inner cavity **211**, the side shell **217** is provided with the air inlet **212** and the debris outlet **213**, at least one of the top portion **215**, the bottom portion **216** and the side shell **217** is provided with the air guide portion **214**.

(31) The top portion **215** is a top shell structure portion of the debris bin housing **21**, and the bottom portion **216** is a bottom shell structure portion of the debris bin housing **21**. A gap is provided between the top portion **215** and the bottom portion **216**. The side shell **217** is enclosed on the peripheral sides of the top portion **215** and the bottom portion **216**, so that the side shell **217**, the top portion **215** and the bottom portion **216** are enclosed to form the inner cavity **211**.

(32) The top portion **215** of the debris bin housing **21** is provided therein with an air outlet channel **218**. The air outlet channel **218** and the inner cavity **211** are arranged in sequence in the height direction of the debris bin housing **21**, the opposite direction of the top portion **215** and the bottom portion **216** is the height direction of the debris bin housing **21**. The air outlet channel **218** is in communication between the inner cavity **211** and the air outlet **219**.

(33) In the embodiment, the filter **40** can be mounted in the air outlet channel **218** inside the debris bin housing **21**, and the top portion **215** includes a top shell body **41** and a movable cover plate **42**, and the top shell body **41** is covered on the side of the side shell **217** away from the bottom portion **216**. The upper surface of the top shell body **41** away from the inner cavity **211** is provided with a groove **43**, and the movable cover plate **42** is rotatably connected to the top shell body **41** to cover or open the groove **43**. The groove **43** forms a part of the air outlet channel **218**. The filter **40** is detachably mounted in the groove **43**, and the filter **40** can be removed and replaced by opening the movable cover plate **42**. In an embodiment, the bottom portion **216** of the groove **43** is provided with a hollow structure, so that the air outlet channel **218** can be arranged in pneumatic

communication with the inner cavity **211** through the hollow structure, and the filter **40** covers the hollow structure to filter debris. The filter **40** may include HEPA or sponge or other porous filter media. In other embodiments, the filter **40** can also be mounted at the air outlet **219**.

(34) In some embodiments, any one of the top portion **215**, the bottom portion **216** and the side shell **217** is provided with the air guide portion **214**, that is, the top portion **215** is provided with the air guide portion **214**, or the bottom portion **216** is provided with the air guide portion **214**, or the side shell **217** is provided with the air guide portion **214**. The number of the air guide portions **214** can be adjusted according to actual needs, and the number of the air guide portions **214** can be one, two or more, which is not limited herein.

(35) In other embodiments, any two of the top portion **215**, the bottom portion **216** and the side shell **217** are provided with the air guide portion **214**, that is, both the top portion **215** and the bottom portion **216** are provided with the air guide portion **214**, or both the top portion **215** and the side shell **217** are provided with the air guide portion **214**; or, both the bottom portion **216** and the side shell **217** are provided with the air guide portion **214**. The number of the air guide portions **214** can be adjusted according to actual needs, and the number of the air guide portions **214** can be one, two or more, which is not limited herein.

(36) In other embodiments, the top portion **215**, the bottom portion **216** and the side shell **217** are all provided with the air guide portion **214**. The number of the air guide portions **214** can be adjusted according to actual needs, and the number of the air guide portions **214** can be one, two or more, which is not limited herein.

(37) It can be understood that the shape of the air guide portion **214** is not limited, and the air guide portion **214** can be in the shape of a rectangle, a triangle, a circle, an ellipse, or a special shape, etc. The size of the cross-sectional area of the air guide portion **214** is not limited. The air guide portion **214** is integrally provided with the debris bin housing **21**, or the air guide portion **214** is a detachable structure on the debris bin housing **21**.

(38) Referring to FIG. 4 and FIG. 5, in an embodiment, the air inlet direction of at least one of the air guide portions **214** is directed toward the bottom portion **216** of the debris bin housing **21**, so that the debris discharging airflow introduced from the air guide portion **214** blows toward the bottom portion **216** of the debris bin housing **21**, which is convenient to lift up the debris in the bottom portion **216** of the debris bin housing **21**, so as to promote the debris (such as hair, large solid objects or wet debris, etc.) that is easily retained in the bottom portion **216** of the debris bin housing **21** flows to the debris outlet **213** following the debris discharging airflow. The air inlet direction of at least one air guide portion **214** disposed on the top portion **215** can be directed to the bottom portion **216** of the debris bin housing **21**, or/and, the air inlet direction of at least one air guide portion **214** disposed on the side shell **217** can be directed to the bottom portion **216** of the debris bin housing **21**, or/and, the air inlet direction of at least one air guide portion **214** disposed on the bottom portion **216** can be directed to the bottom portion **216** of the debris bin housing **21**.

(39) In an embodiment, the air inlet direction of the air guide portion **214** can be arranged perpendicular to the plane where the bottom portion **216** of the debris bin housing **21** is located, or the air inlet direction of the air guide portion **214** can be arranged inclining to the plane where the bottom portion **216** of the debris bin housing **21** is located.

(40) Referring to FIG. 4 to FIG. 7, in an embodiment, the bottom portion **216** is provided with two side regions **51** disposed opposite to each other and a middle region **52** located between the two side regions **51**, and the air inlet **212** of the side shell **217** is arranged corresponding to the middle region **52**, the debris bin housing **21** is provided with a first air guide portion **53** and a second air guide portion **54**, and the first air guide portion **53** and the second air guide portion **54** are respectively provided corresponding to the two side regions **51**, so that the air inlet directions of the first air guide portion **53** and the second air guide portion **54** are directed to the two side regions **51** respectively.

(41) In an embodiment, the debris bin housing **21** has a largest width in the direction from the left

side to the right side, the two side regions **51** are adjacent to the left side and the right side of the debris bin housing **21** respectively, and the middle region **52** is approximately located at the middle position of the debris bin housing **21**. The dust out airflows introduced from the first air guide portion **53** and the second air guide portion **54** are used to flow to the two side regions **51** respectively, so as to drive the debris in the two side regions **51** to flow to the middle region **52**, and then under the action of the vacuum suction at the debris outlet **213** to be discharged from the debris outlet **213**, so that the residual debris in the side regions **51** can be effectively reduced, therefore the debris evacuation efficiency of the debris bin is improved, and the user does not need to manually operate the debris bin to dump out the residual debris. In other embodiments, the top portion **215** can also be provided with one air guide portion **214** or more than two air guide portions **214**. For example, the number of the first air guide portion **53** can be one or two or more, the number of the second air guide portion **54** can be one or two or more.

(42) Referring to FIG. 4 to FIG. 7, in an embodiment, the side shell **217** is provided with a first side wall **2171** and a second side wall **2172** arranged opposite to the first side wall **2171**, and two guide side walls **2173** arranged opposite to each other. The air inlet **212** is arranged penetrating the first side wall **2171**, the debris outlet **213** is arranged penetrating the second side wall **2172**, and the two guide side walls **2173** are respectively recessed toward the left and right sides of the debris bin assembly **20** to form the two side regions **51**, and the two guide side walls **2173** are configured to guide airflow to the debris outlet **213**.

(43) In the embodiment, the two guide side walls **2173** are respectively recessed toward the left and right sides of the debris bin assembly **20** to form the two side regions **51**, so that the debris bin housing **21** has a larger volume to accommodate debris. The two guide side walls **2173** are generally concave curved surfaces, and the distance between the two guide side walls **2173** is narrowed in the direction close to the debris outlet **213**. The two guide side walls **2173** are respectively adjacent to the left and right sides of the debris outlet **213**, and the first air guide portion **53** and the second air guide portion **54** are respectively adjacent to the two guide side walls **2173**, the airflows guided from the first air guide portion **53** and the second air guide portion **54** is directed to the debris outlet **213** in a concentrated manner, thereby the debris concentrated in the two side regions **51** is driven to be directed to the debris outlet **213**, to improve debris evacuation efficiency. In other embodiments, the shapes of the two guide side walls **2173** are not limited to a concave curved surface, and those skilled in the art can set the two guide side walls **2173** to other shapes, such as inclined surfaces, according to actual needs.

(44) The two guide side walls **2173** include a first guide side wall **2173** adjacent to the first air guide portion **53** and a second guide side wall **2173** adjacent to the second air guide portion **54**. The first guide side wall **2173** and the second guide side wall **2173** are adjacent to the left and right sides of the debris outlet **213**, respectively. The relative direction of the first guide side wall **2173** and the second guide side wall **2173** is defined as the width direction of the debris bin housing **21**.

(45) The distance between the air guide portion **214** and the bottom portion **216** is ranged from 40 mm to 50 mm.

(46) The minimum distance $d1$ between the first air guide portion **53** and the first guide side wall **2173** in the width direction of the debris bin housing **21** is ranged from 0 mm to 10 mm, and the minimum distance $d2$ between the first air guide portion **53** and the air inlet **212** in the width direction of the debris bin housing **21** is ranged from 20 mm to 30 mm. The distance $h1$ between the first air guide portion **53** and the bottom portion **216** is ranged from 40 mm to 50 mm. The inventor creative designs the structure of the first air guide portion **53** and the position size of the first air guide portion **53**, so that the height of the starting position, the impact direction and the range of the introduction area of the debris discharging airflow introduced by the first air guide portion **53** can meet the requirements of completely guiding the debris in the side regions **51** to the middle region **52** and then discharge the debris to the debris outlet **213**.

(47) The minimum distance $d2$ between the second air guide portion **54** and the second guide side

wall **2173** in the width direction of the debris bin housing **21** is ranged from 0 mm to 10 mm, and the minimum distance **d4** between the second air guide portion **54** and the air inlet **212** in the width direction of the debris bin housing **21** is ranged from 20 mm to 30 mm. The distance **h2** between the second air guide portion **54** and the bottom portion **216** is ranged from 40 mm to 50 mm. The inventor creative designs the structure of the second air guide portion **54** and the position size of the second air guide portion **54**, so that the height of the starting position, the impact direction and the range of the introduction area of the debris discharging airflow introduced by the second air guide portion **54** can meet the requirements of completely guiding the debris in the side regions **51** to the middle region **52** and then discharge the debris to the debris outlet **213**.

(48) Referring to FIG. 4 to FIG. 7, in an embodiment, the debris outlet **213** and the air inlet **212** are respectively located on opposite sides of the side shell **217**, and the connecting line direction defined between the debris outlet **213** and the air inlet **212** is arranged through the middle region **52**. In the embodiment, the debris outlet **213** and the air inlet **212** are respectively located on the front and rear sides of the debris bin housing **21**. In other embodiments, the debris outlet **213** is generally located on the left or right side of the debris bin housing **21**, or the debris outlet **213** is located on the bottom portion **216** of the debris bin housing **21**.

(49) Referring to FIG. 4 to FIG. 7, in an embodiment, the top portion **215** is provided with an air outlet channel **218** in communication with the inner cavity **211** and an air outlet **219** in communication with the air outlet channel **218**, and the debris bin assembly **20** is provided with a filter **40** located in the air outlet channel **218** to filter the solid particles entering the air outlet channel **218** from the inner cavity **211** through the filter **40**, and the air outlet **219** is used to be arranged in communication with the air suction channel of the cleaning robot **100**, the air inlet **212** is inclined toward the position where the filter **40** is located, the top portion **215** is provided with the air guide portion **214**, and the air guide portion **214** is located on one side of the filter **40** and is staggered from the air inlet direction of air inlet **212**.

(50) In the embodiment, the top portion **215** includes a top shell body **41** and a movable cover plate **42**, and the top shell body **41** is covered on the side of the side shell **217** away from the bottom portion **216**. The upper surface of the top shell body **41** away from the inner cavity **211** is provided with a groove, and the movable cover plate **42** is rotatably connected to the top shell body **41** to cover or open the groove **43**. The groove **43** forms a part of the air outlet channel **218**. The filter **40** is detachably mounted in the groove **43**, and the filter **40** can be removed and replaced by opening the movable cover plate **42**. In an embodiment, the bottom portion **216** of the groove **43** is provided with a hollow structure, so that the air outlet channel **218** can be arranged in pneumatic communication with the inner cavity **211** through the hollow structure, and the filter **40** covers the hollow structure to filter debris.

(51) The number of the air guide portion **214** can be one, two or more, which is not limited herein. In the embodiment, the top portion **215** is provided with a first air guide portion **53** and a second air guide portion **54** opposite to the first air guide portion **53**. The first air guide portion **53** and the second air guide portion **54** are respectively disposed corresponding to the two side regions **51**, that is, the first air guide portion **53** and the second air guide portion **54** are arranged respectively close to the left and right sides of the debris bin housing **21**. The filter **40** is located between the first air guide portion **53** and the second air guide portion **54**, and the filter **40** is disposed corresponding to the middle region **52**. The air inlet direction of the air inlet **212** is arranged through the middle region **52**, so that the air inlet direction of the air inlet **212** is staggered from both of the first air guide portion **53** and the second air guide portion **54**, so that the positions of the first air guide portion **53**, the second air guide portion **54** and the air inlet **212** are not the same. When the debris outlet **213** is subjected to a vacuum suction negative pressure, the air outlet airflow introduced from the air inlet **212** flows to the middle region **52**, and the debris discharging airflow introduced from the first air guide portion **53** and the debris discharging airflow introduced from the second air guide portion **54** respectively flow to the two side regions **51**, due that the positions of the at least

one air guide portion **214** and the air inlet **212** are different, and the at least one air guide portion **214** and the air inlet **212** can guide the debris discharging airflow in different directions, which can increase airflow at the different positions inside the debris bin, which is beneficial to facilitate the discharge of the residual debris in the inner cavity **211** through the debris outlet **213**, therefore the debris evacuation efficiency of the debris bin assembly **20** is improved, and the problem of more debris remaining in the debris bin is solved. In other embodiments, the number of the air guide portion **214** can also be one or more than two.

(52) Referring to FIG. 4 to FIG. 8, in some embodiments, the air guide portion **214** includes an air guide cavity **2141** and a noise reduction assembly **2142**, the air guide cavity **2141** is used to communicate the inner cavity **211** and the external air. The noise reduction assembly **2142** is disposed in the air guide cavity **2141**, and the noise reduction assembly **2142** has a noise reduction effect on the airflow passing through the air guide cavity **2141**. The noise reduction assembly **2142** includes muffler cotton **2143** and a muffler cotton cover **2144**, the muffler cotton **2143** are accommodated in the air guide cavity **2141**, and the muffler cotton cover **2144** covers the air inlet end of the air guide cavity **2141**.

(53) In another embodiment, the air guide portion **214** includes an air guide port and a movable valve. The movable valve is movably disposed at the air guide port to open or cover the air guide port, and the movable valve can open the air guide port in response to the vacuum suction negative pressure at the debris outlet **213**, to introduce the debris discharging airflow through the air guide port, the movable valve covers the air guide port after the vacuum suction negative pressure at the debris outlet **213** is withdrawn. The structure of the movable valve is similar to the structure of the previous debris discharge valve, and reference can be made to the previous description, which is not repeated here. When the cleaning robot is working, the dust suction fan of the cleaning robot generates a first negative pressure in the inner cavity of the debris bin assembly, so as to suck in dust and debris through the air inlet, and the debris discharge valve and the movable valve are all closed; when the cleaning robot is docked with the cleaning base station, the dust suction fan of the cleaning base station generates a second negative pressure in the inner cavity of the debris bin assembly through the debris discharge port, so as to discharge the dust and debris through the debris discharge port, and both the debris discharge valve and the movable valve are in response to the second negative pressure to be in a closed state. Similarly, the noise reduction assembly **2142** can be arranged to reduce the noise of the debris discharging airflow passing through the air guide port, so as to reduce noise.

(54) In another embodiment, the air guide portion **214** includes a plurality of ventilation micro-holes, and the plurality of ventilation micro-holes can function to communicate the inner cavity **211** and the external air, and when the debris outlet **213** is applied to the vacuum suction negative pressure, the plurality of ventilation micro-holes can also introduce the debris discharging airflow to the inner cavity **211**. Similarly, the noise reduction assembly **2142** can be arranged to reduce the noise of the debris discharging airflow passing through the ventilation micro-holes, so as to reduce noise.

(55) Referring to FIG. 8 and FIG. 9, the air guide portion **214** is positioned at the top portion **215** and extends toward the bottom portion **216** to form a protruding portion. The distance between the protruding end of the air guide portion **214** and the bottom portion **216** is ranged from 40 mm to 50 mm. The air guide portion **214** is provided with an air guide channel **2147**, and the air guide channel **2147** extends outward from the protruding end of the protruding portion **2146** to penetrate the outer wall of the debris bin housing **21**, the air guide channel **2147** is in a shape of a horn, and the cross-sectional area of the air guide channel **2147** gradually decreases in the extending direction near the protruding end.

(56) In the embodiment, the debris bin housing **21** is provided with a protruding portion **2146** and an air guide channel **2147** penetrating the protruding portion **2146**, the protruding portion **2146** is protrudingly disposed on the inner wall of the inner cavity **211**, and the air guide channel **2147**

extends outward from the protruding end of the protruding portion **2146** to penetrate the outer wall of the debris bin housing **21**, and the air guide channel **2147** forms the air guide cavity **2141**. In the embodiment, the protruding portion **2146** extends from the top portion **215** toward the bottom portion **216**, and the distance between the protruding end of the protruding portion **2146** and the bottom portion **216** is ranged from 40 mm to 50 mm, so that the protruding portion **2146** is relatively close to the bottom portion **216**, and the protruding portion **2146** increases the length of the air guide channel **2147**, so that the air guide channel **2147** has a sufficient length to guide the debris discharging airflow, which is convenient for guiding the debris discharging airflow oriented to the bottom portion **216** of the debris bin housing **21**, it is convenient to lift up the debris in the bottom portion **216** of the debris bin housing **21**, so as to promote the debris (such as hair, large particle solid objects or wet debris, etc.) that is easily retained in the bottom portion **216** of the debris bin housing **21** to follow the debris discharging airflow to the debris outlet **213**.

(57) Referring to FIG. 1, FIG. 2 and FIG. 11, an embodiment of the present application further provides a cleaning robot system **1000**, where the cleaning robot system **1000** includes the cleaning robot **100** and the cleaning base station **200** described above. The cleaning base station **200** is provided with a debris collection port **50**, and the debris collection port **50** is configured to align with the debris outlet **213** of the debris bin housing **21**. The cleaning base station **200** evacuates the debris from the inner cavity **211** through the debris collection port **50** and the debris outlet **213**.

(58) The cleaning base station **200** includes a base station body **51**, a debris collection container **52** and a suction device **53**. The base station body **51** is provided with a dust collection cavity **54**, a debris collection port **50**, a debris inlet channel **56** and an air outlet channel **55**. The debris collection port **50** is configured to align with the debris outlet **213** of the cleaning robot **100**. One end of the debris inlet channel **30** is connected to the debris collection port **50** and the other end is connected to the dust collection cavity **54**. One end of the air outlet channel **55** is connected to the dust collection cavity **54** and the other end is in communication with the atmosphere. The debris collection container **52** is detachably connected to the dust collection cavity **54** of the base station body **51**, and the debris collection container **52** is mounted in the dust collection cavity **54** to communicate with the debris inlet channel **30** and communicate with the air outlet channel **55**. The suction device **53** is fixed on the base station body **51**, and the suction device **53** is used to drive the airflow to flow.

(59) The suction device **53** can suck the air in the debris collection container **52** and the debris inlet channel **30** through the air outlet channel **55** to generate negative pressure in the debris collection container **52** and the debris inlet channel **30**, so as to evacuate the debris from the debris bin assembly **20** into the debris collection container **52** under the action of negative pressure. In an embodiment, the debris collection container **52** can be a dust collection bag, a dust collection box or a dust collection tank.

(60) It should be noted that the above embodiments are only used to illustrate the technical solutions of the present application, but not to limit herein; under the idea of the present application, the technical features in the above embodiments or different embodiments can also be combined, The steps may be carried out in any order, and there are many other variations of the different aspects of the present application as described above, for brevity, they are not provided in the details; although the present application has been described in detail with reference to the foregoing embodiments, those skilled in the art should understand that: it is still possible to modify the technical solutions described in the foregoing embodiments, or perform equivalent replacements to some of the technical features; and these modification or replacement does not make the essence of the corresponding technical solution deviate from the scope of the technical solutions of the embodiments of the present application.

Claims

1. A debris bin assembly, configured for use with a cleaning robot, and comprising: a debris bin housing, comprising: an inner cavity, an air inlet disposed in communication with the inner cavity and configured for being in communication with a debris inlet channel of the cleaning robot, a debris outlet disposed in communication with the inner cavity, and at least one air guide portion, disposed in communication with the inner cavity; and a debris discharge valve disposed at the debris outlet and configured to open the debris outlet to evacuate debris from the inner cavity in response to a vacuum negative pressure, and to close the debris outlet after the vacuum negative pressure is withdrawn; wherein the at least one air guide portion is configured for being in communication with an external air and configured to guide a debris discharging airflow into the inner cavity in response to the vacuum negative pressure in the debris outlet; wherein the air guide portion comprises an air guide cavity and a noise reduction assembly, and the air guide cavity is configured for being in communication with the inner cavity and the external air, and the noise reduction assembly is arranged in the air guide cavity.
2. The debris bin assembly according to claim 1, wherein the debris bin housing comprises a top portion, a bottom portion disposed opposite to the top portion, and a side shell connected between the top portion and the bottom portion; the top portion, the bottom portion, and the side shell are enclosed to form the inner cavity, the side shell is provided with the air inlet and the debris outlet, and at least one of the top portion, the bottom portion, and the side shell is provided with the at least one air guide portion.
3. The debris bin assembly according to claim 2, wherein an air inlet direction of the at least one air guide portion is directed to the bottom portion of the debris bin housing.
4. The debris bin assembly according to claim 2, wherein the bottom portion is provided with two side regions arranged opposite to each other, and a middle region located between the two side regions, the air inlet of the side shell is arranged corresponding to the middle region, the at least one air guide portion comprises a first air guide portion and a second air guide portion, and the first air guide portion and the second air guide portion are respectively arranged corresponding to the two side regions, such that air inlet directions of the first air guide portion and the second air guide portion are respectively directed to the two side regions.
5. The debris bin assembly according to claim 4, wherein the debris outlet and the air inlet are respectively located on opposite sides of the side shell, and a connecting line direction defined between the debris outlet and the air inlet is arranged passing through the middle region.
6. The debris bin assembly according to claim 4, wherein the side shell is provided with a first side wall, a second side wall arranged opposite to the first side wall, and two guide side walls arranged opposite to each other; the air inlet is arranged penetrating the first side wall, the debris outlet is arranged penetrating the second side wall, and the two guide side walls are respectively recessed towards a left side and a right side of the debris bin assembly to form the two side regions, the two guide side walls are configured to guide airflows to the debris outlet.
7. The debris bin assembly according to claim 2, wherein the top portion is provided with an air outlet channel in communication with the inner cavity and an air outlet in communication with the air outlet channel, and the debris bin assembly is provided in a filter assembly arranged in the air outlet channel to filter solid particles entering the air outlet channel from the inner cavity through the filter assembly, and the air outlet is configured for being in communication with an air suction channel of the cleaning robot, the air outlet is inclined toward a position of the filter assembly, the top portion is provided with the air guide portion, and the air guide portion is located on one side of the filter assembly and is arranged staggered from an air inlet direction of the air inlet.
8. The debris bin assembly according to claim 1, wherein the noise reduction assembly comprises muffler cotton and a muffler cotton cover, the muffler cotton is accommodated in the air guide cavity, and the muffler cotton cover covers an air inlet end of the air guide cavity.
9. The debris bin assembly according to claim 1, wherein the debris bin housing is provided with a

protruding portion and an air guide channel penetrating the protruding portion, and the protruding portion is protrudingly arranged on an inner wall of the inner cavity, the air guide channel extends outward from a protruding end of the protruding portion to penetrate an outer wall of the debris bin housing, and the air guide channel forms the air guide cavity.

10. The debris bin assembly according to claim 2, wherein the air guide portion is positioned at the top portion and extends toward the bottom portion to form a protruding portion.

11. The debris bin assembly according to claim 10, wherein a distance between a protruding end of the air guide portion and the bottom portion is ranged from 40 mm to 50 mm.

12. The debris bin assembly according to claim 10, wherein the air guide portion is provided with an air guide channel, and the air guide channel extends outward from a protruding end of the protruding portion to penetrate an outer wall of the debris bin housing, the air guide channel is in a shape of a horn, and a cross-sectional area of the air guide channel gradually decreases in an extending direction close to the protruding end.

13. The debris bin assembly according to claim 2, wherein the air guide portion includes an air guide port and a movable valve, and the movable valve is movably disposed at the air guide port to open or cover the air guide port, and the movable valve can open the air guide port in response to the vacuum suction negative pressure at the debris outlet, and the movable valve covers the air guide port after the vacuum suction negative pressure at the debris outlet is withdrawn.

14. The debris bin assembly according to claim 2, wherein the side shell is provided with a first side wall, a second side wall disposed opposite to the first side wall, and two guide side walls disposed opposite to each other, the air inlet is arranged penetrating the first side wall, the debris outlet is arranged penetrating the second side wall, the two guide side walls are respectively recessed towards a left side and a right side of the debris bin assembly, the two guide side walls are configured for guiding an airflow to the debris outlet.

15. The debris bin assembly according to claim 14, wherein an opposite direction of the two guide side walls is defined as a width direction of the debris bin housing, the two guide side walls comprises a first guide side wall and a second guide side wall, the at least one air guide portion comprises a first air guide portion and a second air guide portion, the first air guide portion and the second air guide portion are respectively arranged adjacent to the first guide side wall and the second guide side wall.

16. The debris bin assembly according to claim 15, wherein a minimum distance between the first air guide portion and the first guide side wall in the width direction of the debris bin housing is ranged from 0 mm to 10 mm; a minimum distance between the first air guide portion and the air inlet in the width direction of the debris bin housing is ranged from 20 mm to 30 mm.

17. The debris bin assembly according to claim 16, wherein a minimum distance between the second air guide portion and the second guide side wall in the width direction of the debris bin housing is ranged from 0 mm to 10 mm; a minimum distance between the second air guide portion and the air inlet in the width direction of the debris bin housing is ranged from 20 mm to 30 mm.

18. A cleaning robot, comprising a robot body and a debris bin assembly, wherein the debris bin assembly comprises: a debris bin housing, comprising: an inner cavity, an air inlet disposed in communication with the inner cavity and configured for being in communication with a debris inlet channel of the cleaning robot, a debris outlet disposed in communication with the inner cavity, and at least one air guide portion, disposed in communication with the inner cavity; and a debris discharge valve disposed at the debris outlet and configured to open the debris outlet to evacuate debris from the inner cavity in response to a vacuum negative pressure, and to close the debris outlet after the vacuum negative pressure is withdrawn; wherein the at least one air guide portion is configured for being in communication with an external air and configured to guide a debris discharging airflow into the inner cavity in response to the vacuum negative pressure in the debris outlet; wherein the debris bin assembly is detachably mounted on the robot body; and wherein the air guide portion comprises an air guide cavity and a noise reduction assembly, and the air guide

cavity is configured for being in communication with the inner cavity and the external air, and the noise reduction assembly is arranged in the air guide cavity.

19. A debris bin assembly, applied to a cleaning robot, and comprising: a debris bin housing, comprising: an inner cavity, an air inlet disposed in communication with the inner cavity and configured for being in communication with a debris inlet channel of the cleaning robot, a debris outlet disposed in communication with the inner cavity, and at least one air guide portion, disposed in communication with the inner cavity; and a debris discharge valve disposed at the debris outlet and configured to open the debris outlet to evacuate debris from the inner cavity in response to a vacuum negative pressure, and to close the debris outlet after the vacuum negative pressure is withdrawn; wherein the at least one air guide portion is configured for being in communication with an external air and configured to guide a debris discharging airflow into the inner cavity in response to the vacuum negative pressure in the debris outlet; wherein the debris bin housing comprises a top portion, a bottom portion disposed opposite to the top portion, and a side shell connected between the top portion and the bottom portion; the top portion, the bottom portion, and the side shell are enclosed to form the inner cavity, the side shell is provided with the air inlet and the debris outlet, and at least one of the top portion, the bottom portion, and the side shell is provided with the air guide portion; wherein the air guide portion includes an air guide port and a movable valve, and the movable valve is movably disposed at the air guide port to open or cover the air guide port, and the movable valve can open the air guide port in response to the vacuum suction negative pressure at the debris outlet, and the movable valve covers the air guide port after the vacuum suction negative pressure at the debris outlet is withdrawn.
